

6

Elementary Logic

*... if it was so, it might be; and if it were so, it would be:
but as it isn't, it ain't. That's logic.*

— Tweedledee in *Through the Looking-Glass*
by Lewis Carroll (1832–1898)

Why is it important to study logic? Two things that we continually try to accomplish are to understand and to be understood. We attempt to understand an argument given by someone so that we can agree with the conclusion or, possibly, so that we can say that the reasoning does not make sense. We also attempt to express arguments to others without making a mistake. A formal study of logic will help improve these fundamental communication skills.

Why should a student of computer science study logic? A computer scientist needs logical skills to argue whether or not a problem can be solved on a machine, to transform logical statements from everyday language to a variety of computer languages, to argue that a program is correct, and to argue that a program is efficient. Computers are constructed from logic devices and are programmed in a logical fashion. Computer scientists must be able to understand and apply new ideas and techniques for programming, many of which require a knowledge of the formal aspects of logic.

In this chapter we'll discuss the formal character of sentences that contain words like "and," "or," and "not" or a phrase like "if A then B ."

Chapter Guide

Section 6.1 starts our study of logic with the philosophical question "How do we reason?" We'll discuss some common things that we all do when we reason, and we'll introduce the general idea of a "calculus" as a thing with which to study logic.